

WHELKS ARE LIKE LITTLE WOLVES IN SLOW MOTION

KEYSTONE SPECIES

IN CONTEXT

KEY FIGURE

Robert Paine (1933–2016)

BEFORE

1950s In Kenya, farmer and conservationist David Sheldrick introduces elephants to Tsavo East National Park, and discovers this results in a major increase in biodiversity.

1961 Fieldwork by American ecologist Joseph Connell on Scotland's rocky shores shows that removing predatory whelks alters the distribution of their barnacle prey.

AFTER

1994 In the US, a group of ecologists led by Brian Miller publishes a paper explaining the valuable role prairie dogs play as a keystone species.

2016 Fieldwork leads marine ecologist Sarah Gravem to conclude that organisms can be keystone species in some places but not in others.

A keystone species plays a crucial role in the way an ecosystem functions, even though it is often a small part of the overall biomass of the ecosystem. Because it exerts a disproportionately large effect on the environment relative to its biomass, if a keystone species disappears from an ecosystem,